Sandeep Satheesh

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Education

University of Illinois, Urbana-Champaign — Urbana-Champaign, Illinois
Master of Computer Science — *2026 – 2028*

Indian Institute of Technology, Madras — Chennai, India
Bachelor of Science (B.Sc) - Programming & Data Science; Concurrent Degree; GPA: 3.6/4.0 — *2022 – 2026*

Anna University — Chennai, India
Bachelor of Engineering (B.E) - Electronics and Communication Engineering; GPA: 3.73/4.0 — *2021 – 2025*

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL
Backend: FastAPI, Django, React, PostgreSQL, MongoDB, Redis, Elasticsearch, PydanticAI, LangChain, PyTorch
Infrastructure & Tools: Docker, Kubernetes, AWS, GCP, Azure DevOps, Git, Kafka, Celery, Prometheus, Grafana

Professional Experience

Disability Resources & Educational Services (DRES), UIUC — Urbana-Champaign, Illinois
Graduate Intern — *Aug 2026 – Present*

- Developing OCR-based document processing and automated note-generation pipelines, integrating model inference, validation, and workflow automation to accelerate accessibility document preparation and reduce manual processing.

S2T AI — Singapore
Software Engineer L1 — *Sept 2025 – Aug 2026*

- Built and deployed **∼20 Python/FastAPI microservices** for automated data-ingestion and backend workflows, using **Kafka, Celery, Redis, MongoDB, and Elasticsearch** for asynchronous processing, caching, persistence, and search.
- Maintained **200+ production APIs** serving internal and external customers and handling **∼40K requests/day**, while supporting ingestion systems processing **∼100K new data sources/day**.
- Built internal failure-tracking and notification tooling for ingestion pipelines, helping sustain **∼99.8% reliability** across systems processing **∼100K new data sources/day**, with **Prometheus/Grafana** observability and **Docker/Kubernetes** deployment.

Vessel Match (Start-up at Nirmaan IIT Madras) — Chennai, India
Founding Engineer — *May 2024 – Aug 2025*

- Architected backend APIs and **orchestration workflows** for maritime vessel–cargo matching, integrating private/public vessel-tracking APIs with prospective-client data sources.
- Built **structured tool interfaces** for agents to retrieve, validate, and normalize vessel, cargo, and availability data before executing matching workflows.
- Reduced vessel-matching turnaround from **hours of manual email/search work to minutes**, while processing **200+ vessels/day** and improving match relevance for cargo and customer preferences.

Indian Institute of Technology, Madras — Chennai, India
Teaching Assistant - Business Data Management (MS2001) — *Jan 2024 – Sept 2025*

- Built Python + LLM evaluation scripts to automate submission checks and curate datasets, improving grading speed and data integrity while supporting student doubt-clearing sessions.

Projects

ReverserX [Code] — *Aug 2026 – Present*
Built a standalone Android reverse-engineering platform integrating **JADX, ADB, Frida, mitmproxy, and Ghidra** for static analysis, runtime instrumentation, network inspection, and native-code analysis. Implemented safe APK/APKM ingestion, source indexing, structured evidence collection, and reproducible CLI workflows.
Indexed **155K+ source files** into **917K+ searchable chunks**; implemented **SQLite/ChromaDB** persistence, checkpoints, failure recovery, bounded execution, and evidence-linked reporting, with **232 passing tests** and **85% test coverage**.
Python · JADX · ADB · Frida · mitmproxy · Ghidra · SQLite · ChromaDB

GuruHeal [Presentation Code] — *May 2025*
Built a multi-service healthcare platform with **FastAPI** and **Next.js/TypeScript**, using PostgreSQL/Supabase for persistence and Redis for caching, with streaming conversations, document retrieval, external search, multilingual support, and asynchronous backend processing.
FastAPI · Next.js · TypeScript · PostgreSQL · Redis · Supabase · Docker

Research & Publications

Knee Osteoarthritis Detection & Grading *(IEEE CSNT '25; Indian Patent)* [Paper] — *2024 – 2025*

Built a multimodal **EfficientNetB3 + XGBoost** pipeline for KOA detection, grading, interpretation, and reporting.

Leadership, Honors & Awards

- **Winner — Karnataka State Police Datathon, 2024**: Won among 30,000+ submissions; built a geospatial and time-series crime analysis system for Karnataka State Police with support from Microsoft.
- **Paradox, IIT Madras**: Secretary, 2025; Steering Committee, 2026 — led 14 departments and 50+ events for 5,000+ attendees, overseeing logistics, financial audits, and an INR 16M budget.